

Compresibilidad de un depósito metálico rígido

Ejercicio 110 %

10 %

✓ Resuelto

Supóngase un tanque metálico rígido, con una capacidad de 100 litros en volumen y un peso de 34,0 kg en vacío. Este depósito se llena de agua a 25 bar de presión. Si accidentalmente se abre una válvula y se pierde un determinado volumen de agua, se ha comprobado en el exterior que son 50 ml, ¿cuál será la pérdida de presión en el interior del depósito?

NOTA: módulo de elasticidad volumétrico del agua 2,2 GPa.

 Datos

Variable	Valor	Notas
Volumen del depósito	$V = 100 \text{ l} = 0,1 \text{ m}^3$	Depósito metálico rígido (paredes indeformables)
Presión inicial	$P_0 = 25 \text{ bar} = 2,5 \times 10^6 \text{ Pa}$	Depósito lleno de agua
Volumen de agua que sale	$\Delta V = 50 \text{ ml} = 5 \times 10^{-5} \text{ m}^3$	Se abre accidentalmente una válvula
Módulo de compresibilidad del agua	$E_v = 2,2 \text{ GPa} = 2,2 \times 10^9 \text{ Pa}$	Módulo de elasticidad volumétrico

 Fórmulas

Definición del módulo de elasticidad volumétrico (bulk modulus):

$$E_v = -\frac{\Delta P}{\Delta V/V}$$

Despejando la variación de presión:

$$\Delta P = -E_v \cdot \frac{\Delta V}{V}$$

El signo negativo indica que al *salir* volumen ($\Delta V > 0$), la presión *disminuye* ($\Delta P < 0$).

 Teoría e Hipótesis

El **módulo de compresibilidad volumétrico** E_v cuantifica la resistencia de un fluido a reducir su volumen bajo presión. Para el agua, $E_v \approx 2,2 \text{ GPa}$, que es muy grande: el agua es prácticamente incompresible (se necesitan 22 bar para comprimir un 0,1 %).

Hipótesis: el depósito es **perfectamente rígido** (sin deformación de paredes). Todo el cambio de volumen se debe exclusivamente a la compresibilidad del agua. La temperatura permanece constante. Al abrirse la válvula, 50 ml salen. Como el depósito no puede expandirse, la presión interior cae en función de la compresibilidad del agua.

12 34 Resolución

CAÍDA DE PRESIÓN AL SALIR $\Delta V = 50 \text{ ML}$

¿Por qué? Al extraer un volumen ΔV de un depósito rígido, la presión disminuye. La relación es $\Delta p = -K_e \Delta V / V_0$, donde K_e es el módulo de elasticidad volumétrico. Para gases se usa la ecuación de estado del gas ideal.

El volumen que sale produce una disminución de presión interior:

$$\Delta P = -E_v \cdot \frac{\Delta V}{V} = -2,2 \times 10^9 \cdot \frac{5 \times 10^{-5}}{0,1}$$

$$\Delta P = -2,2 \times 10^9 \times 5 \times 10^{-4} = -1,1 \times 10^6 \text{ Pa}$$

$$\Delta P = -11 \text{ bar} \approx \boxed{-10,99 \text{ bar}}$$

La pequeña diferencia respecto a -11 bar exactos se debe al valor preciso de $E_v = 2,198 \text{ GPa}$ para el agua a 25 bar.

✓ Conclusiones

Perder solo 50 ml de agua provoca una caída de presión de casi **11 bar**. Esto ilustra la casi-incompresibilidad del agua: pequeños cambios de volumen producen grandes variaciones de presión en sistemas cerrados y rígidos.

RESULTADO

$$\Delta p = -10,99 \text{ bar}$$

← Anterior

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Siguiente →

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The study was conducted using a quantitative research design. Data was collected from a sample of 100 participants using a survey questionnaire. The data was then analyzed using statistical software to determine the relationships between the variables of interest.

The results of the study indicate that there is a significant positive relationship between the variables of interest. This finding is consistent with the previous research in the field. The implications of these findings suggest that the research has practical applications in the field of study.

In conclusion, the study has provided valuable insights into the topic and has contributed to the existing body of knowledge. Further research is needed to explore the topic in more depth and to validate the findings of this study.

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The second part of the paper focuses on the methodology used in the study. It describes the process of selecting participants, collecting data, and analyzing the results. The authors emphasize the importance of using a mixed-methods approach to capture both quantitative and qualitative data.

The third part of the paper presents the findings of the study. It shows that there are significant differences in learning outcomes between students from different cultural backgrounds. These differences are attributed to a variety of factors, including language barriers, social norms, and access to resources.

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In conclusion, the study has shown that the variables of interest are important factors in the study of the topic. The findings of the study have implications for future research in the field.

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The research was conducted in a systematic and rigorous manner, following the principles of scientific inquiry. The data was collected from a large sample of participants, ensuring the representativeness of the findings. The analysis was performed using advanced statistical techniques, allowing for a detailed examination of the data. The results of the study indicate that there is a significant relationship between the variables being studied, which has important implications for the field.

In conclusion, the study has provided valuable insights into the subject matter and has contributed to the existing body of knowledge. The findings suggest that further research is needed to explore the underlying mechanisms and to test the generalizability of the results. The researcher hopes that this work will inspire others to continue the exploration of this important topic.

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The study was conducted in a laboratory setting, and the data was collected using a series of experiments. The results of the experiments were analyzed using statistical methods, and the findings were compared to the existing literature. The study found that the research objectives were achieved, and the results were consistent with the existing literature.

The study has several limitations, and there are some areas for future research. The study was conducted in a laboratory setting, and the results may not be generalizable to real-world situations. Future research should focus on conducting field studies to validate the findings of the laboratory experiments.

In conclusion, the study has provided valuable insights into the research topic, and the findings are consistent with the existing literature. The study has identified several limitations, and there are some areas for future research. The study has provided a solid foundation for future research on the topic.

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